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By Amit Katiyar (MCA-JNU)



MATHEMATICS

- The number of 5 people groups that can be selected from 9 people when two particular persons are not to be in same group is
(a) 126 (b) 35
(c) 91 (d) 252
- The solution set of equation $\log_x 2 \log_{2x} 2 = \log_{4x} 2$ is
(a) $\{2^{-\sqrt{2}}, 2^{\sqrt{2}}\}$ (b) $\{\frac{1}{2}, 2\}$
(c) $\{\frac{1}{4}, 2^2\}$ (d) $\{\frac{1}{4}, 2\}$
- If twelve sides regular polygon is inscribed in a circle of radius 3 centimeters, then length of each side of the polygon is
(a) $\frac{3\sqrt{6}-3\sqrt{2}}{2}$ (b) $18 - 9\sqrt{3}$
(c) $18 + 9\sqrt{3}$ (d) none
- If C is the mid-point of AB and P is any point outside AB, then
(a) $PA + PB = 2PC$ (b) $PA + PB = PC$
(c) $PA + PB + 2PC = 0$ (d) $PA + PB + PC = 0$
- The average marks of boys in class is 52 and that of girls to 42. The average marks of boys and girls combined is 50, then percentage of boys in the class is
(a) 80% (b) 60%
(c) 40% (d) 20%
- A box contains 2 blue caps, 4 red caps, 5 green caps and 1 yellow cap. If four caps are picked at random, then the probability that none of them is green, is
(a) $\frac{7}{99}$ (b) $\frac{7}{12}$
(c) $\frac{5}{99}$ (d) $\frac{5}{12}$
- The line $3x + 5y = k$ touches the ellipse $16x^2 + 25y^2 = 400$ if k is
(a) $\pm\sqrt{5}$ (b) $\pm\sqrt{15}$
(c) ± 25 (d) $\pm\sqrt{35}$
- If $X = \{4^n - 3n - 1, n \in N\}$ and $Y = \{9n - 9, n \in N\}$, then $X \cup Y$ is equal to
(a) y (b) X
(c) N (d) None of these
- $\int \left\{ \frac{(\log x - 1)}{1 + (\log x)^2} \right\}^2 dx$ is equal to
(a) $\frac{xe^x}{1+x^2} + C$ (b) $\frac{x}{(\log x)^2 + 1} + C$
(c) $\frac{\log x}{(\log x)^2 + 1} + C$ (d) $\frac{x}{x^2 + 1} + C$
- The volume of the parallelopiped determined $U = \hat{i} + 2\hat{j} - \hat{k}, V = -2\hat{i} + 3\hat{k}$ and $W = 7\hat{j} - 4\hat{k}$ is
(a) 21 (b) 22
(c) 23 (d) 24
- The vector perpendicular to the plane passing through $(1, -1, 0), (-2, 1, -1)$ and $(-1, 1, 2)$ is
(a) $6\hat{i} + 6\hat{j}$ (b) $6\hat{i} + 7\hat{k}$
(c) $7\hat{i} + 6\hat{k}$ (d) none
- The equation of a circle with diameters are $2x - 3y + 12 = 0$ and $x + 4y - 5 = 0$ and area of 154 sq units is
(a) $x^2 + y^2 - 6x + 4y - 36 = 0$
(b) $x^2 + y^2 + 6x - 4y - 36 = 0$
(c) $x^2 + y^2 - 6x - 4y + 25 = 0$
(d) None of the above
- $\int \frac{x^2 - 1}{x^3 \sqrt{2x^4 - 2x^2 + 1}} dx$ is equal to
(a) $\frac{\sqrt{2x^4 - 2x^2 + 1}}{x^2} + C$ (b) $\frac{\sqrt{2x^4 - 2x^2 + 1}}{x^3} + C$
(c) $\frac{\sqrt{2x^4 - 2x^2 + 1}}{x} + C$ (d) $\frac{\sqrt{2x^4 - 2x^2 + 1}}{2x^2} + C$
- If a, b and a+b are vectors of magnitude a, then the magnitude of vector a-b is
(a) $\sqrt{2}a$ (b) $\sqrt{3}a$
(c) 2a (d) 3a
- Which of the following statements if FALSE?
(a) $2 \in A \cup B$ implies that if $2 \notin A$, then $2 \in B$
(b) $[2, 3] \subseteq A$ implies that $2 \subseteq A$ and $[2, 3] \subseteq B$
(c) $A \cup B \supseteq [2, 3]$ implies that $[2, 3] \subseteq A$ and $[2, 3] \subseteq B$
(d) $[2] \in A$ and $[3] \in B$ implies that $[2, 3] \subseteq A$
- If $2x^2 + 7xy + 3y^2 + 8x + 14y + \lambda = 0$ represents pair of straight lines, the value of λ is
(a) 2 (b) 4
(c) 6 (d) 8
- The area of the region bounded by the lines $y = |x - y|$ and $y = 3 - |x|$ is
(a) 3 sq units (b) 4 sq units
(c) 6 sq units (d) 2 sq units
- In a triangle ABC, $a = 4, b = 3, \angle BAC = 60^\circ$, then the equation for which c is the root, is
(a) $c^2 + 3c + 7 = 0$ (b) $c^2 + 3c - 7 = 0$
(c) $c^2 - 3c + 7 = 0$ (d) $c^2 - 3c - 7 = 0$
- If $\cos \theta = \frac{5}{13}, \frac{3\pi}{2} < \theta < 2\pi$, then 2θ is
(a) $\frac{-120}{119}$ (b) $\frac{-120}{169}$
(c) $\frac{119}{116}$ (d) $\frac{120}{119}$
- An expression has 10 equally likely outcomes. Let A and B two non-empty events of the experiment. If consists of 4 outcomes, the number of outcomes that B must have so that A and B are independent, is
(a) 2, 4, or 8 (b) 3, 6 or 9
(c) 4 or 8 (d) 5 or 10
- Let a, b and b be three non-zero vectors, no two of which are collinear. If the vector $a + 2b$ is collinear with c and $b + 3c$ is collinear with a, then $a + 2b + 6c$ is equal to
(a) a (b) b
(c) c (d) 0
- The value of a, for which the sum of the squares of the roots of the equation $x^2 - (a - 2)x - (a + 1) = 0$, assumes the least value is
(a) 3 (b) 2
(c) 0 (d) 1



23. For any two events A and B, the probability that atleast one of them occur is 0.6. If A and B occurs simultaneously with a probability 0.3, then $P(A') + P(B')$ is
(a) 0.9 (b) 1.15 (c) 1.1 (d) 1.0
24. Two finite sets A and B are having m and n elements. The total numbers subsets of the first set is 56 more than the total number of subsets of the second set. The value of m and n are
(a) 7, 6 (b) 6, 3 (c) 5, 3 (d) 8, 7
25. The probability that A speaks truth is $\frac{4}{5}$ while this probability for B is $\frac{3}{4}$. the probability that they contradict each other when asked to speak on a fact is
(a) $\frac{3}{20}$ (b) $\frac{1}{5}$ (c) $\frac{7}{20}$ (d) $\frac{4}{5}$
26. The sum of the expression $\frac{1}{\sqrt{1+\sqrt{2}}} + \frac{1}{\sqrt{2+\sqrt{3}}} + \frac{1}{\sqrt{3+\sqrt{4}}} + \dots + \frac{1}{\sqrt{80+81}}$ is
(a) 7 (b) 8 (c) 9 (d) 10
27. Consider the function f defined by $f(x) = \begin{cases} x^2 - 1, & x < 3 \\ 2ax, & x \geq 3 \end{cases}$ for all real number x. If f is continuous at $x = 3$, then value a is
(a) 8 (b) $\frac{3}{4}$ (c) $\frac{1}{8}$ (d) $\frac{4}{3}$
28. Three houses are available in a locality. Three persons apply for the houses. Each applies for one house without consulting others. The probability that all the three apply for the same house is
(a) $\frac{8}{9}$ (b) $\frac{7}{9}$ (c) $\frac{2}{9}$ (d) $\frac{1}{9}$
29. Five horses are in a race. Mr. A selects two of the horses at random and bets on them. The probability that Mr. A selected the winning horse is
(a) $\frac{3}{5}$ (b) $\frac{1}{5}$ (c) $\frac{2}{5}$ (d) $\frac{4}{5}$
30. If $3^x = 4^{x-1}$, then x is equal to
(a) $\frac{2-\log_3 2}{2\log_3 2-1}$ (b) $\frac{2}{2\log_3 2-1}$ (c) $\frac{2-\log_3 2}{2\log_3 2+1}$ (d) $\frac{2\log_3 2}{2\log_3 2-1}$
31. The matrix A has x rows and (x + 5) columns and the matrix D has y rows and (11 - y) Columns. If both the matrices AB and BA exist, then the value of x and y are
(a) 8, 3 (b) 3, 5 (c) 3, 8 (d) 8, 5
32. A circus artist is climbing a 20 m long rope, which is tightly stretched and tied from the top of a vertical pole to the ground. Find the height of the pole, if the angle made by the rope with the ground level is 30° .
(a) 10m (b) 20m (c) 30m (d) 40m
33. There are n equally speed points 1, 2, ..., n marked on the circumference of a circle. If the point 15 is directly opposite of the point 49, then the total number of points is
(a) 50 (b) 68 (c) 66 (d) 70
34. Let $S = \{1, 2, \dots, n\}$. The number of possible pairs of form (A, B) with $A \subseteq B$ for subsets A, B of S is
(a) 2^n (b) 3^n (c) n! (d) none
35. Sum of the roots of the equations $4^x - 3(2^{x+3}) + 128 = 0$ is
(a) 5 (b) 6 (c) 7 (d) 8
36. If the sum of the slopes of the lines given by $x^2 - 2cxy - 7y^2 = 0$ is four times their product, then the value of c is
(a) 1 (b) -1 (c) -2 (d) 2
37. The system of equations $x + y + 2z = a$
 $x + z = b$
 $2x + y + 3z = c$ has a solution if
(a) $B = c$ (b) $c = a + b$
(c) $c = a + 2b$ (d) $a = b = c$
38. Let $f(x) = x^2 - bx + c$, b is an odd positive integer. If $f(x) = 0$ has two prime number as roots and $b + c = 35$, then the global minimum value of $f(x)$ is
(a) $-\frac{183}{4}$ (b) $\frac{173}{16}$
(c) $-\frac{81}{4}$ (d) $\frac{17}{2}$
39. The vertex of the parabola whose focus is (-1, 1) and directrix is $4x + 3y - 24 = 0$ is
(a) $(0, \frac{3}{2})$ (b) $(0, \frac{5}{2})$
(c) $(1, \frac{3}{2})$ (d) $(1, \frac{5}{2})$
40. The number of points in $(-\infty, \infty)$, for which $x^2 - x \sin x - \cos x = 0$ is
(a) 6 (b) 4 (c) 2 (d) 0
41. There are 4 books on fairy tales, 5 novels and 3 plays. In how many ways can they be arranged in the order, books on fairy tales, novels and then plays so that the books of the same category are put together?
(a) 17280 (b) 103680 (c) 51840 (d) 360
42. Suppose a population A has 100 observations 101, 102, ..., 200 and another population B has 100 observation 151, 152, ..., 250. If V_A and V_B represent variance of the two population respectively, then $\frac{V_A}{V_B}$ is
(a) $\frac{9}{4}$ (b) $\frac{4}{9}$ (c) 1 (d) $\frac{2}{3}$
43. If a, b are vectors such that $|a + b| = \sqrt{29}$ and $a \times (2\hat{i} + 3\hat{j} + 4\hat{k}) = (2\hat{i} + 3\hat{j} + 4\hat{k}) \times b$, then a possible value of $(a + b) \cdot (-7\hat{i} + 2\hat{j} + 3\hat{k})$ is
(a) 0 (b) 3 (c) 4 (d) 8
44. Let $x_1, x_2, \dots, x_n$ Be n observation such that $\sum x_i^2 = 400$ and $\sum x = 80$. Then a possible value of n among the following is
(a) 10 (b) 15 (c) 20 (d) 8
45. Area the greatest rectangle that can be inscribed in the ellipse is
(a) $\sqrt{ab}$ (b) $2ab$ (c) ab (d) $\frac{a}{b}$
46. Two common tangents to the circle $x^2 + y^2 = 2a^2$ and parabola $y^2 = 8ax$ are
(a) $x = \pm(y + 2a)$ (b) $y = \pm(x + 2a)$
(c) $x = \pm(y + a)$ (d) $y = \pm(x + a)$



47. If $a_1, a_2, \dots, a_n$ are in AP and $a_1 = 0$, then the value $\left(\frac{a_3}{a_2} + \frac{a_4}{a_3} + \dots + \frac{a_n}{a_{n-1}}\right) - a_2 \left(\frac{1}{a_2} + \frac{1}{a_3} + \dots + \frac{1}{a_{n-2}}\right)$ equal to
- (a) $(n-2) + \frac{1}{(n-2)}$ (b) $\frac{1}{(n-2)}$
 (c) $(n-2)$ (d) $n - \frac{1}{(n-2)}$
48. The value of $\cos 20^\circ + \cos 100^\circ + \cos 140^\circ$ is
- (a) 0 (b) $\frac{1}{\sqrt{2}}$
 (c) $\frac{1}{2}$ (d) 1
49. The permutations of (a, b, c, d, e, f, g) are listed in lexicographic order. Which of the following permutation are just before and just after the permutation bacdefg?
- (a) agefedbc and bacdfge
 (b) agfedcb and badcefg
 (c) agfebcd and bacedgf
 (d) agfedcb and bacdegf
50. The foci the ellipse $\frac{x^2}{16} + \frac{y^2}{b^2} = 1$ and the hyperbola $\frac{x^2}{144} - \frac{y^2}{81} = 1$ coincide. The the value of b^2 is
- (a) 5 (b) 7
 (c) 9 (d) 1

LOGICAL REASONING

Directions (Q. Nos. 51-52) Are based on the following A boy is asked to put in a basket one mango when ordered 'One', one orange when ordered 'Two'. One apple when ordered 'Three' and is asked to take out from the basket one mango and an orange when ordered 'Four'. A sequence of order is given as

12332142314223314113234

51. How many total fruits will be in the basket at the end of the above order sequence?
- (a) 9 (b) 8
 (c) 11 (d) 10
52. How many total oranges were in the basket at the end of the above the sequence?
- (a) 1 (b) 4
 (c) 3 (d) 2

Directions (Q. Nos. 53 to 56) : Are based on the following: Eight friends J, K, L, M, N, O, P and Q live on eight different floors of a building but not necessarily in same order. The lower most floor of the building is number one, the above that number two and soon until the top most floor is number eight.

- J lives on floor number six.
- Only one person lives between J and L.
- O lives on the floor immediately below L.
- Only one person lives between O and P.
- O lives above P.
- K lives on an even number floor but not on the floor number two.
- Two persons live between K and Q.
- Q does not live on the lower most floor
- N lives on one of the floor above Q.

53. If P and L Interchange their places, who will live between P and M?
- (a) O (b) L (c) J (d) No one
54. Which of the following is True about 'M'?
- (a) K lives immediately above M
 (b) Only two people live between M and Q
 (c) M lives on an odd numbered floor
 (d) M lives on the lower most floor
55. Who amongst the following lives on the floor number eight?
- (a) P (b) O
 (c) K (d) Cannot be determined
56. Three of the following four are alike in a certain way based on the given arrangement and thus form a group. Which of the following does not belong to the group?
- (a) PL (b) MQ
 (c) LN (d) OM

Directions (Q. Nos. 57 to 59): Are based on the following A team must be selected from the ten probable players A, B, C, D, E, F, G, H, I and J. Of these, A, C, E and J are forwards, B, G and H are point guards and D, F and I are defender.

- The team must have at least one forward, one point guard and one defender.
 - If the team include J, it must also include F.
 - The team must include E or B, but not both.
 - If the team include G, it must also include F.
 - The team must include exactly one among C, G and I.
 - C and F cannot be members of the same team.
 - D and H cannot be members of the same team.
 - The team must include both A and D or neither of them.
 - There is no restriction on the number of members in the team.
57. What would be the size of the largest possible team?
- (a) 7 (b) 6 (c) 5 (d) 4
58. Which of the following cannot be included in a team of size 6?
- (a) A (b) H (c) J (d) E
59. What could be the maximum size of the team that include G?
- (a) 4 (b) 5
 (c) 6 (d) More than 6
60. A family has several children. Each boy in this family has as many sisters as brothers, but each girls has twice as many brothers as sister. How many brothers and sister are there?
- (a) 1 and 2 (b) 3 and 4
 (c) 6 and 3 (d) 4 and 3
61. In a certain code language '134' means 'Good and tasty', '478' means 'see good picture' and '729' means 'picture are faint'. Which of the following numerical symbols stand for see?
- (a) 2 (b) 7 (c) 8 (d) 1



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62. If the English word 'EXAMINATION' is coded as 56149512965, then word 'GOVERNMENT' can be coded as
 (a) 7655955552 (b) 7645954452
 (c) 7645954552 (d) 7644956552
63. **Fact 1 :** Most stuffed toys are stuffed with beans.
Fact 2 : There are stuffed bears and stuffed tigers.
Fact 3 : Some chair are stuffed with beans.
 If the above statement are fact. Which of the following statements, must also be fact?
 1. Only children's chairs are stuffed with beans.
 2. All stuffed tigers are stuffed with beans.
 3. Stuffed monkeys are not stuffed with beans.
 (a) 1 is fact
 (b) Only 2 is fact
 (c) Both 2 and 3 are facts
 (d) None of the statements 1, 2, 3 are true
64. If all the 6's are replaced by 9's then the algebraic sum of all the numbers from 1 to 100 (both inclusive) varies by
 (a) 333 (b) 300 (c) 279 (d) 330
65. 405 sweets were distributed equally among a group of children such that the number of sweet received by each child is one fifth of the number of children. The number of children in the group is
 (a) 45 (b) 9 (c) 21 (d) 15
66. The number of common terms in the two sequence 17, 21, 25, , 817 and 16, 21, 26, , 851 is
 (a) 28 (b) 39 (c) 40 (d) 87
67. Unscramble the letter of the following words and find the odd one.
 (a) ONGEAR (b) NOONI
 (c) ALPEP (d) AUVAG
68. A bus starts from its depot filled to seating capacity. It stops at a point A where $\frac{1}{6}$ th of passengers alight and 10 board the bus. At point B, $\frac{1}{5}$ th of the passengers alight and boards the bus. At point C which is the last stop, all the 55 passengers alight. The capacity of the bus is
 (a) 96 (b) 99 (c) 66 (d) 90
69. Kha- kha is an obscure island which is inhabited by two types of people; the 'Yes' type and the 'No' type. Native of types 'Yes' ask only questions the right answer in which is 'Yes' while those of type 'No' ask only questions the right answer to which is 'No'. For example the 'Yes' type will ask questions like "Is 2 plus 2 equal to 4?" While the "No" type will ask question like "Is 2 plus 2 equal to 5?". The following question is based on your visit to the island Kha-kha. Kevin and Kumar are brothers from the island. Kumar asks you, Is atleast one of us is of type 'No'? You can conclude that
 (a) Kevin is 'No', Kumar is 'Yes'.
 (b) Both are 'Yes'.
 (c) Kevin is 'Yes', Kumar is 'No'.
 (d) Both are 'No'.

70. Raman was born on March 5, 1970. Lakshman was born 25 days before Raman. The year when they took birth. the Republic Day fell on Monday. What is the day of birth of Lakshman.
 (a) Sunday (b) Monday
 (c) Wednesday (d) Saturday
71. P, Q, R and S are four logical statements such that if P is true, then Q is true; if Q is true, then R is true; and if S is true, then at least one of Q and R is false. The it follows that
 (a) if S is false, then both Q and R true
 (b) if at least one of Q and R is true, then S is false
 (c) if P is true, then S is false
 (d) if Q is true, then S is false
72. A, B, C, D, E, F and G are members of a family consisting of four adults, three children, three males and four female. Out of the children F and G are girls. A and D are brothers and A is doctor, E is an engineer married to one of the brothers and has two children. B is married to D and G is their child. Who is C?
 (a) E's daughter (b) A's son
 (c) G's brother (d) F's father
73. A clock is set right at 5 am the clock losses 16 minutes in 24 hours. What will be the correct time when the clock indicates 10 pm on the 3rd day?
 (a) 11 pm (b) 10: 45 pm
 (c) 11: 15 pm (d) 12 pm
74. The day after the day after tomorrow is four days after Monday. What day is it today?
 (a) Monday (b) Tuesday
 (c) Wednesday (d) Thursday

Directions (Q. Nos. 75 to 77) :

- There are six houses P, Q, R, S, T and U, three on either side of a road.
 - The houses are of different colours - red, blue, green, orange, yellow and white.
 - All the houses are of different heights.
 - T, the tallest house is exactly opposite to the red coloured house.
 - The shortest house is exactly opposite to the green coloured house.
 - U, the orange coloured house is located between P and S.
 - R, the yellow coloured house is exactly opposite to P.
 - Q, the green coloured house is exactly opposite to U.
 - P, the white coloured house is taller than R, but shorter than S and Q.
75. Which is the second largest house?
 (a) Q (b) R
 (c) S (d) Cannot be determined
76. Which of the second shortest houses?
 (a) P (b) R
 (c) S (d) Cannot be determined

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77. What is the colour of the tallest house?

- (a) Red (b) Blue
- (c) Green (d) Yellow

78. How many pairs of letters are there in the word NECESSARY which have as many letters between them in the word as there are between them in the alphabet and in the same order.

- (a) One (b) two
- (c) three (d) four

79. What is missing number in the series, 4, 7, 11, 18, 29, 47,....., 123, 199 ?

- (a) 76 (b) 77 (c) 86 (d) 87

Directions (Q. Nos. 80 to 82) :

- A, B, C, D, E and F are six members in a family in which there are two married couples.
 - D is brother of F.
 - Both D and F are lighter than B.
 - B is mother of D and lighter than E.
 - C a lady, is neither heaviest nor lightest in the family.
 - E is lighter than C.
 - The grandfather in the family is the heaviest.
80. Which of the following is a pair of married couples.
- (a) AB (b) BC
 - (c) AD (d) BE
81. Who among the following will be in the second place, if all the members in the family are arranged in a descending order of their weights?
- (a) C (b) A
 - (c) D (d) Data inadequate
82. How is C related to D?
- (a) Grandmother (b) Cousin
 - (c) Sister (d) Mother

Directions (Q. Nos. 83 to 85) : are based on the following:

- Eleven students, A, B, C, D, E, F, G, H, I, J and K are sitting in the first row of the class facing the teacher.
 - D who is to the immediate left of F is second to the right of C.
 - A is second to the right of E, who is at one of the ends.
 - J is the immediate neighbor of A and B and third to the left of G.
 - H is to the immediate left of D and third to the right of I.
83. Who is sitting in the middle of the row?
- (a) B (b) C (c) G (d) I
84. If E and D, C and B, A and H and K and F interchange their position, which of the following pairs of students are sitting at the ends?
- (a) D and E (b) E and F
 - (c) D and K (d) K and F
85. Which of the following groups of friends is sitting to the right of G?
- (a) CHDE (b) CHDF
 - (c) IBJA (d) None of the above

Directions (Q. Nos. 86 to 89) are based on the following:

- A group of six friends are sitting around a hexagonal table, each one at one corner of the hexagon.
 - Ram is sitting opposite to Ramesh.
 - Jyoti is sitting next to Seema.
 - Neeta is sitting opposite to Seema but not next to Ram.
 - Amrit has a person sitting between Ramesh and himself.
86. If Seema and Jyoti mutually interchange their positions, then who will be sitting opposite to Neeta?
- (a) Jyoti (b) Ram
 - (c) Seema (d) Ramesh
87. If Neeta sits to the right of Amrit, then who is sitting to the left of Amrit?
- (a) Ramesh (b) Neeta
 - (c) Jyoti (d) Ram
88. Who is sitting between Amrit and Ramesh?
- (a) Neeta (b) Jyoti
 - (c) Seema (d) Ram
89. Who is sitting opposite to Jyoti?
- (a) Ramesh (b) Neeta
 - (c) Amrit (d) Seema
90. A Group of 630 children are seated in n rows for a group photo session. Each row contains three less children than the row in front of it. Which one of the following number of rows is not possible?
- (a) 3 (b) 4 (c) 5 (d) 6

GENERAL ENGLISH

Directions (Q. Nos. 91 to 94) are based on the following.

White cement is the basic raw material for producing cement tiles and cement paint which are used extensively in building construction. The main consumers of white cement are, therefore, cement tile and cement paint manufacturing units. These consumers, mostly in the small scale sector, are today facing a major crisis because of a significant increase in the price of white cement during a short period. The present annual licensed production capacity of white and grey cement in a country is approximately 3.5 lakh tones. The average demand is 2-2.5 lakh tones. This means that there is idle capacity to the tune of one lakh tones or more, The price is, therefore, not a phenomenon arising out of inadequate production capacity but evidently because of artificial scarcity created by the manufactures in their self-interest.

The main reason for the continuing spurt in cement price is its decontrol. As it is, there is stiff competition in the cement paint and tile manufacturing business. Any further price revision at this stage is bound to have a severe adverse impact on the market conditions. The Government should take adequate steps to ensure that suitable controls are brought in. Else it should allow import of cement.

91. Which of the following words has the opposite meaning as the word "basic" as used in the passage?
- (a) Vital (b) Unimportant
 - (c) Acidic (d) Last

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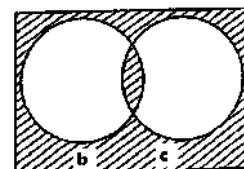
92. Why is the price of cement going up?
 (a) Because the government is controlling quota
 (b) Because of export of white cement
 (c) Because of the large usage of white cement
 (d) None of the above
93. What is the crisis faced by the cement tile manufactures as described in the passage?
 (a) White cement prices are very high
 (b) White cement is not of good quality
 (c) White cement usage is high
 (d) White cement is priced very low
94. Which of the following words has the same meaning as the word 'artificial' as used in the passage?
 (a) Deliberate (b) Prolonged
 (c) Practical (d) Unnatural
95. Choose the words that accurately signifies a person who makes money by starting or running business.
 (a) Antreprenour (b) Andrapreneur
 (c) Entraprenour (d) Entrepreneur
96. Which of the following is correct phrase to describe a group of insects?
 (a) A flock of insects (b) A swarm of insects
 (c) A school of insects (d) A school of insects
97. Which of the following has closest meaning to the word 'REPUTATION'?
 (a) Character (b) Respect
 (c) Fame (d) Honour
98. Which of the following means 'Theatrical' ?
 (a) Thrilling (b) Histrionic
 (c) Delicate (d) Delicious
99. Identify the word which is different from the rest of the words
 (a) Indisputable (b) uncertain
 (c) Dubious (d) Doubtful
100. Select the pair that best expresses a relationship similar to the expressed in SCALE: TONE
 (a) Physician: Medicine (b) Wave: Amplitude
 (c) Spectrum: Colour (d) Rainbow: Shower
101. Choose the answer which best expresses the meaning of the idiom/phrase "to burn a hole in the pocket".
 (a) Steal from someone's pocket
 (b) To destroy other's belongings
 (c) to be very miserly
 (d) Money that is spent quickly
102. Choose the correct alternative to fill the blank. My window look..... the garden.
 (a) An (b) out on (c) in (d) at
103. Fill in the blank with suitable article.darkest cloud has a silver lining
 (a) An (b) A (c) The (d) From
104. Fill in blank with appropriate adjective. The streak is completely..... it is cold and tough.
 (a) edible (b) erratic
 (c) unswerving (d) thedible

Fill in the blank with a suitable preposition.

105. We have been looking for a new flatages.
 (a) since (b) for (c) during (d) in
106. Fill in the blank With appropriate verb Where is he? He should home hours ago.
 (a) be (b) have been
 (c) had been (d) were
107. Fill in the blank With appropriate question tag You shouldn't be here on a holiday,
 (a) Shouldn't you (b) should you not
 (c) would' t you (d) Should you
108. Change the following sentence into passive sentence. They studied Mathematics last year.
 (a) Mathematics was studied by them last year
 (b) Mathematics were studied by them last year
 (c) Mathematics has been studied by them the last year
 (d) Mathematics studied them last year.
109. The meaning of the word "EGRESS" is
 (a) Entrance (b) Exit
 (c) Double (d) Program
110. Choose the answer which best expresses the meaning of the Idiom/phrase " Elbow room"
 (a) Opportunity for freedom of action
 (b) Special room for the guest
 (c) To give enough space to move or work in
 (d) To add a new room to the house

COMPUTER AWARENESS

111. Consider a hard disk with 16 recording surfaces (0-15) having 16384 cylinders (0- 16383) and each cylinder contains 64 sectors (0-63). Data storage capacity in each sector is 512 bytes. Data are organized cylinder- wise and the addressing format is < cylinder no. , surface no., sector no.>. A file of size 42797 KB is stored in the disk and the starting disk location of the file is <1200, 9,40>. What is the cylinder number of the last sector of the file, if it is stored in a continuous manner?
 (a) 1284 (b) 1282
 (c) 1286 (d) 1288
112. Consider the following minterm expression for F.
 $F(P, Q, R, S) = \sum 0, 2, 5, 7, 8, 10, 13, 15$
 The minterms 2, 7, 8 and 13 are 'do not care' terms. The minimum sum- of- product form for F is
 (a) $Q\bar{S} + \bar{Q}S$
 (b) $QS + \bar{Q}\bar{S}$
 (c) $\bar{Q}\bar{R}\bar{S} + \bar{Q}R\bar{S} + Q\bar{R}S + QRS$
 (d) $\bar{P}\bar{Q}\bar{S} + \bar{P}QS + PQS + P\bar{Q}S$
113. The bollean expression represented by the following Venn diagram is
 (a) a XOR b
 (b) $a'b + ab'$
 (c) $ab + a'b$
 (d) $(a + b)(a' + b)$





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114. The range of n-bit signed magnitude representation is
 (a) 0 to $2^n - 1$ (b) $-(2^{n-1} - 1)$ to $(2^{n-1} - 1)$
 (c) $-(2^n - 1)$ to $(2^{n-1} - 1)$ (d) 0 to $2^{n-1} - 1$
115. A hard disk has a rotational speed of 6000 rpm. Its average latency time is
 (a) 5×10^{-3} sec (b) 00.05 sec
 (c) 1 sec (d) 0.5 sec
116. The 2's complement representation of the number $(-100)_{10}$ in an 8-bit computer is
 (a) 10011011 (b) 01100100
 (c) 11100100 (d) 10011100
117. The number of terms in the product of sum canonical form of $|(x_1 + x_2)(x_3x_4)|$ is
 (a) 7 (b) 8
 (c) 9 (d) none of these

118. Find the odd out:
 (a) HTTP
 (b) FCFS
 (c) HTML
 (d) TCP/IP
119. Consider the equation $(43)_x = (y3)_8$ where x and y are unknown. The number of possible solution are
 (a) 4 (b) 6
 (c) 5 (d) 7
120. Subtract $(1010)_2$ from $(1101)_2$ using first complement
 (a) $(1100)_2$
 (b) $(0101)_2$
 (c) $(1001)_2$
 (d) $(0011)_2$

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MAARULA MATHEM -E LEARNING

ANSWER KEY

1.	(c)	2.	(a)	3.	(b)	4.	(a)	5.	(a)
6.	(a)	7.	(c)	8.	(a)	9.	(b)	10.	(c)
11.	(d)	12.	(b)	13.	(d)	14.	(b)	15.	(a)
16.	(d)	17.	(b)	18.	(d)	19.	(d)	20.	(d)
21.	(d)	22.	(d)	23.	(c)	24.	(b)	25.	(c)
26.	(b)	27.	(d)	28.	(d)	29.	(c)	30.	(d)
31.	(c)	32.	(a)	33.	(c)	34.	(d)	35.	(c)
36.	(d)	37.	(b)	38.	(c)	39.	(d)	40.	(d)
41.	(a)	42.	(c)	43.	(c)	44.	(c)	45.	(b)
46.	(b)	47.	(a)	48.	(a)	49.	(d)	50.	(b)
51.	(c)	52.	(d)	53.	(a)	54.	(b)	55.	(c)
56.	(d)	57.	(a)	58.	(b)	59.	(c)	60.	(d)
61.	(c)	62.	(b)	63.	(d)	64.	(d)	65.	(a)
66.	(c)	67.	(b)	68.	(c)	69.	(a)	70.	(a)
71.	(c)	72.	(b)	73.	(b)	74.	(b)	75.	(d)
76.	(b)	77.	(b)	78.	(a)	79.	(a)	80.	(d)
81.	(a)	82.	(a)	83.	(d)	84.	(c)	85.	(b)
86.	(a)	87.	(d)	88.	(a)	89.	(c)	90.	(d)
91.	(b)	92.	(d)	93.	(a)	94.	(a)	95.	(d)
96.	(b)	97.	(c)	98.	(b)	99.	(a)	100.	(c)
101.	(d)	102.	(b)	103.	(c)	104.	(a)	105.	(b)
106.	(b)	107.	(d)	108.	(a)	109.	(b)	110.	(a)
111.	(a)	112.	(b)	113.	(c)	114.	(b)	115.	(a)
116.	(d)	117.	(d)	118.	(b)	119.	(c)	120.	(d)

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SOLUTION

MATHS

1. (c) Total number of peoples = 9

Number of ways in which 5 people groups can be selected from 9 people. According to the question, 5 people are selected from 9-2 particulars people already in the same group.

$$\Rightarrow {}^9C_5 - {}^7C_3 \Rightarrow \frac{9!}{5!4!} - \frac{7!}{3!4!} \Rightarrow \frac{9 \times 8 \times 7 \times 6}{4 \times 3 \times 2} - \frac{7 \times 6 \times 5}{3 \times 2}$$

$$\Rightarrow 126 - 35 \Rightarrow 91$$

2. (a) $x > 0, x \neq 1, \frac{1}{2}, \frac{1}{4}$

$$\frac{\log_2 2 \log_2 2}{\log_2 x \log_2 2x} = \frac{\log_2 2}{\log_2 4x}$$

$$\Rightarrow \frac{1}{\log_2 x (1 + \log_2 x)} = \frac{1}{(2 + \log_2 x)}$$

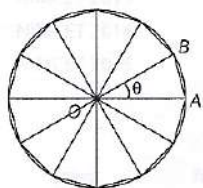
$$\Rightarrow \frac{1}{t} \times \frac{1}{(1+t)} = \frac{1}{2+t}$$

$$\Rightarrow 2+t = t^2 + t \Rightarrow t^2 = 2$$

$$\Rightarrow t = \pm\sqrt{2} \Rightarrow \log_2 x = \sqrt{2} \text{ and } \log_2 x = -\sqrt{2}$$

$$\Rightarrow x = 2^{\sqrt{2}}, 2^{-\sqrt{2}}$$

3. (b) Since, it is a regular polygon (it has 12 sides).



Each side will subtend an equal $\angle\theta$ at centre

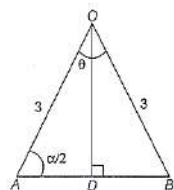
$$12\theta = 360^\circ$$

$$\Rightarrow \theta = 30^\circ$$

$$OA = OB = \text{Radius} = 3 \text{ cm}$$

$$\text{Sum of internal angles of a polygon} = (n-2)180^\circ$$

Let the internal angle be α .



$$\text{Hence, } (12-2) \times 180^\circ = 12\alpha$$

$$\Rightarrow \alpha = 150^\circ$$

$$\therefore \angle A = \angle B = \frac{\alpha}{2} = 75^\circ$$

$$\therefore \cos 75^\circ = \frac{AD}{3}$$

$$\Rightarrow AD = 3\cos 75^\circ$$

$$\therefore AB = 6\cos 75^\circ$$

$$= 6 \times \frac{(\sqrt{3}-1)}{2\sqrt{2}} = \frac{3}{\sqrt{2}}(\sqrt{3}-1) > 0$$

Since, $AB < 3$

$$\frac{3}{\sqrt{2}}(\sqrt{3}-1) < 3$$

Here, exact answer does not match.

From given options only (b) $18 - 9\sqrt{3}$ satisfies the result.

4. Using, triangle law of addition

$$\vec{PA} + \vec{AC} > \vec{PC}$$

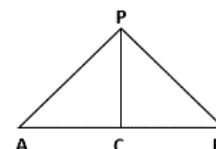
$$\vec{PB} + \vec{BC} > \vec{PC}$$

$$\vec{PA} + \vec{AC} + \vec{PB} + \vec{BC} = 2\vec{PC}$$

$$PA + PB + AC + BC = 2PC$$

$$PA + PB(AC, BC) = 2PC$$

$$\vec{PA} + \vec{PB} = 2\vec{PC} \quad (A)$$



5. Let b and g be the no of boys and girls respectively.

Average marks of boys (X_1) = 52

Average marks of girls (X_2) = 42

Combined average marks of boys and girls (x) = 50

$$x = \frac{bx_1 + gx_2}{b+g}$$

$$50 = \frac{52b + 42g}{b+g}$$

$$8g = 2b \Rightarrow b > 49$$

$$\text{Percentage of boys} = \frac{b}{b+g} \times 100$$

$$= \frac{4g}{5g} \times 100$$

$$= 80\%$$

6. (a) Box contains

2 blue caps, 4 red caps

5 green caps and 1 yellow cap

Total number of caps = 12

Sample space = ${}^{12}C_4$

For favourable cases = 7C_4

(None of them is green = total caps - green caps

$$= 12 - 5 = 7)$$

$$\text{Probability} = \frac{{}^7C_4}{{}^{12}C_4} = \frac{7}{99}$$

7. (c)

Given equation of line is $3x + 5y = k$

Comparing Eq. (i) with $ix + my + n = 0$, we get

$$l = 3, m = 5 \text{ and } n = -k$$

Equations of ellipse is

$$16x^2 + 25y^2 = 400$$

On divide Eq. (ii) by 400, we get

$$\frac{16x^2}{400} + \frac{25y^2}{400} = \frac{400}{400}$$

$$\Rightarrow \frac{x^2}{25} + \frac{y^2}{16} = 1$$

By comparing with standard form of ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

We get $a = 5, b = 4$

$\therefore$ Condition of tangency

$$n^2 = a^2l^2 + b^2m^2$$

$$(-k)^2 = (5)^2(3)^2 + (4)^2(5)^2$$

$$\Rightarrow k^2 = 225 + 400$$

$$\Rightarrow k^2 = 625$$

$$\Rightarrow k = \pm 25$$

8. (a) $X = 4^n - 3n - 1$

$$n = 1 \Rightarrow 4 - 3 - 1 = 0$$

$$n = 2 \Rightarrow 16 - 6 - 1 = 9$$



$$X = \{0, 9, 54, \dots\}$$

Now,

$$Y = 9n - 9 = 9(n - 1)$$

$$n = 1 \Rightarrow 9(1 - 1) = 0$$

$$n = 2 \Rightarrow 9(2 - 1) = 9$$

$$n = 3 \Rightarrow 9(3 - 1) = 18$$

$$Y = \{0, 9, 18, \dots\}$$

$$X \subset Y$$

Hence, $X \cup Y = Y$

9. (b) $\int \frac{\{(\log x - 1)\}^2}{1 + (\log x)^2} dx$

$$= \int \frac{(\log x - 1)^2}{(1 + (\log x)^2)^2} dx = \int \frac{((\log x)^2 + 1) - 2\log x}{(1 + (\log x)^2)^2} dx$$

$$= \int \frac{1}{(1 + (\log x)^2)} - \int \frac{2\log x}{(1 + (\log x)^2)^2}$$

Using integration by parts in first integral,

$$\int (1 + (\log x)^2)^{-1} - \int \frac{2\log x}{(1 + (\log x)^2)^2}$$

$$= x \cdot (1 + (\log x)^2)^{-1} - \int (-1)(1 + (\log x)^2)^{-2} \times \frac{2\log x}{x} \cdot x$$

$$- \int \frac{2\log x}{(1 + (\log x)^2)^2}$$

$$= \frac{x}{(1 + (\log x)^2)} + \int \frac{2\log x}{(1 + (\log x)^2)^2} - \int \frac{2\log x}{(1 + (\log x)^2)^2} + C$$

$$= \frac{x}{1 + (\log x)^2} + C$$

10. (c) Volume of parallelepiped is $[U \ V \ W]$

$$\text{Volume} = \begin{vmatrix} 1 & 2 & -1 \\ -2 & 0 & 3 \\ 0 & 7 & -4 \end{vmatrix} = -23$$

$$\text{Volume} = |-23| = 23 \text{ units}$$

11. None option is correct

$$\text{Let } A = \hat{i} - \hat{j}, B = -2\hat{i} + \hat{j} - \hat{k}, C = -\hat{i} + \hat{j} + 2\hat{k}$$

$$AB = -3\hat{i} + 2\hat{j} - \hat{k}, BC = \hat{i} + 3\hat{k}$$

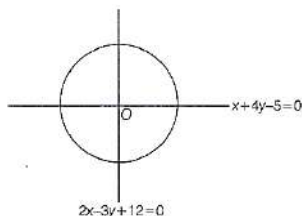
$$AB \times BC = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -3 & 2 & -1 \\ 1 & 0 & 3 \end{vmatrix} = 6\hat{i} + 8\hat{j} + (-2)\hat{k}$$

$$AB \times BC = 6\hat{i} + 8\hat{j} - 2\hat{k}$$

12. (b) Centre is the intersection point of the diameters.

$$\therefore 2x - 3y + 12 = 0$$

$$2x + 8y - 10 = 0$$



Solving these two equations, we get

Centre $O(-3, 2)$

$$\text{Given, } A = \pi r^2 = 154$$

$$\Rightarrow \frac{22}{7} \times r^2 = 154 \Rightarrow r = 7 \text{ units.}$$

$$\therefore \text{Equation, } (x + 3)^2 + (y - 2)^2 = 49$$

$$x^2 + y^2 + 6x - 4y - 36 = 0$$

13. (d) $\int \frac{x^2 - 1}{x^3 \sqrt{2x^4 - 2x^2 + 1}} = \int \frac{x^2 - 1}{x^3 \sqrt{x^4(2 - \frac{2}{x^2} + \frac{1}{x^4})}}$

$$= \int \frac{x^2 - 1}{x^5 \sqrt{2 - \frac{2}{x^2} + \frac{1}{x^4}}}$$

Now, substitute

$$2 - \frac{2}{x^2} + \frac{1}{x^4} = t$$

$$\Rightarrow 2 - 2x^{-2} + x^{-4} = t$$

$$\Rightarrow (0 - 2(-2)x^{-3} - 4x^{-5})dx = dt$$

$$\Rightarrow (4x^{-3} - 4x^{-5})dx = dt \Rightarrow (x^{-3} - x^{-5})dx = \frac{dt}{4}$$

Then,

$$= \int \frac{x^{-3} - x^{-5}}{\sqrt{2 - \frac{2}{x^2} + \frac{1}{x^4}}} = \int \frac{dt/4}{\sqrt{t}} = \frac{1}{4} \int \frac{dt}{\sqrt{t}} = \frac{1}{4} \times 2\sqrt{t}$$

$$= \frac{1}{2} \sqrt{2 - \frac{2}{x^2} + \frac{1}{x^4}} + C = \frac{\sqrt{2x^4 - 2x^2 + 1}}{2x^2} + C$$

14. (b) $|a + b|^2 = |a|^2 + |b|^2 + 2ab$

$$\Rightarrow \alpha^2 = \alpha^2 + \alpha^2 + 2a \cdot b \Rightarrow 2a \cdot b = -\alpha^2$$

$$|a - b|^2 = |a|^2 + |b|^2 - 2a \cdot b$$

$$\Rightarrow |a - b|^2 = \alpha^2 + \alpha^2 + \alpha^2 = 3\alpha^2 \Rightarrow |a - b| = \sqrt{3}\alpha$$

15. (a)

Question have two options right which is b and

(a)

$2 \in A \cup B$ implies that if $2 \notin A$, then $2 \in B$ It is clearly right.

(b) $\{2, 3\} \subseteq A$ implies that $2 \subseteq A$ and $\{2, 3\} \subseteq B$ Clearly, it is wrong.

(c) $A \cup B \supseteq \{2, 3\}$ implies that $\{2, 3\} \subseteq A$ and $\{2, 3\} \subseteq B$ Clearly, it is wrong.

(d) $\{2\} \in A$ and $\{3\} \in A$ implies that $\{2, 3\} \subseteq A$

Ex. $A = \{1, 2, 3, 4\}, \{2\} \in A$ and $\{3\} \in A$

Implies $\{2, 3\} \subseteq A$

It is clearly right.

16. (d) $2x^2 + 7xy + 3y^2 + 8x + 14y + \lambda = 0$

For pair of straight line $\Delta = 0$

$$abc + 2fgh - af^2 - bg^2 - ch^2 = \theta$$

$$a = 2, b = 3, h = \frac{7}{2}, g = 4, f = 7, c = \lambda$$

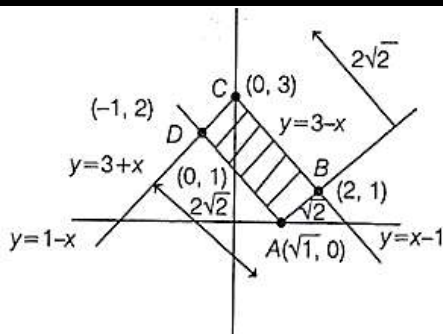
$$6\lambda + 2 \times \frac{7}{2} \times 4 \times 7 - (2 \times 49) - (3 \times 16) - \left(\lambda \times \frac{49}{4}\right) = 0$$

$$\Rightarrow 6\lambda + 196 - 146 - \frac{49\lambda}{4} = 0 \Rightarrow \frac{24\lambda - 49\lambda}{4} + 50 = 0$$

$$\Rightarrow -25\lambda = -200 \quad \lambda = 8$$

17. (d) $y = |x - 1| = \begin{cases} x - 1; & x \geq 1 \\ -x + 1; & x < 1 \end{cases}$

$$y = 3 - |x| = \begin{cases} 3 - x; & x \geq 0 \\ 3 + x; & x < 0 \end{cases}$$



As AD is parallel to BC and AB is parallel to CD

Here, $AB = CD = \sqrt{2}$
 $BC = AD = 2\sqrt{2}$

and

Area of rectangle $ABCD = l \times b$
 $= 2\sqrt{2} \times \sqrt{2}$
 $= 4$ sq. units

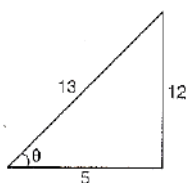
18. (d) $\cos A = \frac{b^2 + c^2 - a^2}{2bc} \Rightarrow \cos 60^\circ = \frac{9 + c^2 - 16}{6c}$

$$\frac{1}{2} = \frac{c^2 - 7}{6c}$$

$$\Rightarrow c^2 - 3c - 7 = 0$$

19. (d)

$$\cos \theta = \frac{5}{13}, \frac{3\pi}{2} < \theta < 2\pi \Rightarrow 3\pi < 2\theta < 4\pi$$



$$\Rightarrow \tan \theta = \frac{12}{5}$$

$$\because \theta \in 4^{\text{th}} \text{ quad} \Rightarrow \tan \theta = \frac{-12}{5}$$

$$\Rightarrow \tan 2\theta = \frac{2\tan \theta}{1 - \tan^2 \theta}$$

$$= \frac{2\left(\frac{-12}{5}\right)}{1 - \left(\frac{-12}{5}\right)^2} = \frac{-24(5)}{25 - 144}$$

$$= \frac{-120}{-119} = \frac{120}{119}$$

20. (d) Let $n(A)$, $n(B)$ and $n(A \cap B)$ denote the number outcomes favourable to A , B and $(A \cap B)$ respectively. If A and B are independent.

$$P(A \cap B) = P(A)P(B)$$

$$\Rightarrow \frac{n(A \cap B)}{10} = \frac{4}{10} \times \frac{n(B)}{10}$$

$$\Rightarrow 5 \times n(A \cap B) = 2 \times n(B)$$

$$\text{Hence, } n(B) = \frac{5}{2} n(A \cap B)$$

For independent events $n(A \cap B) \geq 1$

For different values of $n(A \cap B) = 2, 4$ as $n(B)$ must be a natural number so $n(B) = 5, 10$

As, we know $n(B) \leq 10$.

21. (d) $a + 2b = \lambda c$... (i)

$b + 3c = \mu a$... (ii)

Eq. (i) $- 2 \times$ Eq. (ii)

$$a - 6c = \lambda c - 2\mu a \Rightarrow (1 + 2\mu)a = (\lambda + 6)c$$

As we know that a and c is non collinear vectors,

Hence, But no two are collinear, so it means

$$\lambda + 6 = 0 \Rightarrow \lambda = -6$$

$$1 + 2\mu = 0 \Rightarrow \mu = -\frac{1}{2}$$

Put these values in Eq. (i)

$$a + 2b = \lambda c$$

$$a + 2b = -6c$$

$$a + 2b + 6c = 0$$

22. (d): $\alpha + \beta = a - 2; \alpha\beta = -(a + 1)$

$$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = (a - 2)^2 + 2(a + 1)$$

$$= a^2 + 4 - 4a + 2a + 2 \Rightarrow f(a) = a^2 - 2a + 6$$

$$\text{For least value } f'(a) = 0 \Rightarrow \frac{d}{da}(a^2 - 2a + 6) = 0$$

$$\Rightarrow 2a - 2 = 0 \Rightarrow a = 1$$

23. (c) We are given that $P(A \cup B) = 0.6$

$$\text{and } P(A \cap B) = 0.3$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$0.6 = (1 - P(\bar{A})) + (1 - P(\bar{B})) - 0.3$$

$$P(\bar{A}) + P(\bar{B}) = 2 - 0.3 - 0.6 = 1.1$$

24. (b) According to question, $2^m - 2^n = 56$.

For $m = 6$, $n = 3$ above satisfies, so $m = 6$, $n = 3$.

25. (c) The probability of speaking truth of A is $P(A) = \frac{4}{5}$ The

probability A speaks lie $P(\bar{A}) = 1 - \frac{4}{5} = \frac{1}{5}$ The probability of

speaking truth of B is $P(B) = \frac{3}{4}$ The probability B speaks lie

$P(\bar{B}) = 1 - \frac{3}{4} = \frac{1}{4}$ The probability of they contradict each other

$$= P(A) \times P(\bar{B}) + P(\bar{A}) \times P(B)$$

$$= \frac{4}{5} \times \frac{1}{4} + \frac{1}{5} \times \frac{3}{4} = \frac{1}{5} + \frac{3}{20} = \frac{7}{20}$$

26. $\frac{1}{\sqrt{1}+\sqrt{2}} + \frac{1}{\sqrt{2}+\sqrt{3}} + \frac{1}{\sqrt{3}+\sqrt{4}} + \dots \dots \dots \frac{1}{\sqrt{80}+\sqrt{81}}$

$$S = \frac{\sqrt{2}-\sqrt{1}}{2-1} + \frac{\sqrt{3}-\sqrt{2}}{3-2} + \dots \dots \dots \frac{\sqrt{81}-\sqrt{80}}{81-80}$$

$$S = \sqrt{2} - \sqrt{1} + \sqrt{3} - \sqrt{2} + \dots \dots + \sqrt{81}$$

$$S = 9 - 1$$

$$S = 8 \text{ (B).}$$

27. (d) $\text{LHL} = \lim_{x \rightarrow 3^-} x^2 - 1$

$$= 9 - 1 = 8$$

$$\text{RHL} = \lim_{x \rightarrow 3^+} 2a \cdot 3 = 6a$$

Function is $x \rightarrow 3^+$ continuous at $x = 3$

Hence, $\text{LHL} = \text{RHL}$

$$\Rightarrow 8 = 6a$$

$$\Rightarrow a = \frac{4}{3}$$



28. (d) No. of houses = 3 = No. of favourable cases

No. of applicants = 3

Total number of events = 3^3

(because each candidate can apply in 3 ways)

Required probability = $\frac{1}{3^3} + \frac{1}{3^3} + \frac{1}{3^3} = \frac{3}{3^3} = \frac{1}{9}$

29. (c) In total there are 5 horses, out of these Mr. A selects two horse. This can be done in ${}^5C_2 = 10$ ways.

Out of the five horses, only one horse wins.

Number of ways to select winning horse = 1

Number of ways to select one other horse out of 4 horses

= ${}^4C_1 = 4$

The probability that Mr. A selected the winning horse is

= $\frac{4}{10} = \frac{2}{5}$

30. (d) Given, $3^x = 4^{x-1}$

Taking log both sides of base 3,

$$\log_3 3^x = \log_3 4^{x-1}$$

$$\Rightarrow x = (x-1)\log_3 4 \Rightarrow x = (x-1)\log_3 2^2$$

$$\Rightarrow x = 2(x-1)\log_3 2$$

$$\Rightarrow x = (2x-2)\log_3 2 \Rightarrow x = 2x\log_3 2 - 2\log_3 2$$

$$\Rightarrow 2\log_3 2 = 2x\log_3 2 - x$$

$$\Rightarrow 2\log_3 2 = x(2\log_3 2 - 1) \Rightarrow x = \frac{2\log_3 2}{(2\log_3 2 - 1)}$$

31. (c)

If AB possible, then number. of column in A = Number of rows in B

$$x+5 = y \Rightarrow x-y = -5$$

If BA possible, the then number of column in B = Number of rows in A

$$11-y = x$$

$$\Rightarrow x+y = 11$$

Solve both Eqs. (i) and (ii)

$$x-y = -5 \quad 2x = 6 \quad 3-y = -5$$

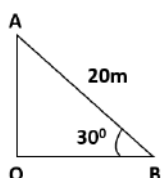
$$x+y = 11 \quad x = 3 \quad y = 8$$

32. (a)

$$\sin 30^\circ = \frac{OA}{AB}$$

$$\frac{1}{2} = \frac{OA}{20}$$

$$OA = 10m$$



33. (c)

There are given n equally spaced points 1,2,...,n

Given 15 is directly opposite to the point 49

So other point will be $49 - 15 = 34$

But two points must be exclude,

So, there are 32 points which apposite

Then, total number of points are

$$= 32 \times 2 + 2$$

$$= 64 + 2 = 66$$

34. (*) Given, $S = \{1, 2, \dots, n\}$

First chose a k element subset B of S. This can be done in $\binom{n}{k}$ ways.

Given the subset B, A has $2^k - 1$ options (since the numbers of proper subset of a k - element set is $2^k - 1$)

Hence, the total number of pairs of sets (A, B) such that $A \subseteq B$ is given by

$$\sum_{k=0}^n (2^k - 1) \binom{n}{k} = \sum_{k=0}^n \binom{n}{k} 2^k - \sum_{k=0}^n \binom{n}{k}$$

$$= 3^n - 2^n$$

None option is correct.

$$35. (c) (2^x)^2 - 3(2^x \cdot 2^3) + 2^7 = 0$$

$$\text{Let } 2^x = t > 0$$

$$\Rightarrow t^2 - 3 \cdot 2^3 t + 2^7 = 0 \Rightarrow t^2 - 24t + 128 = 0$$

$$\Rightarrow (t-16)(t-8) = 0 \Rightarrow t = 16, 8 \Rightarrow 2^x = 2^4, 2^3$$

$$\Rightarrow x = 4, 3 \Rightarrow \text{Sum of roots} = 4 + 3 = 7$$

36. (d) Let m_1 and m_2 be the slope of lines

$$x^2 - 2xyc - 7y^2 = 0$$

$$m_1 + m_2 = \frac{-2n}{b}$$

$$m_1 + m_2 = \frac{-2c}{7}$$

$$m_1 + m_2 = \frac{a}{b} = \frac{-1}{7}$$

$$m_1 + m_2 \Rightarrow 4m_1 + m_2$$

$$\frac{-2c}{7} = 4 \times \frac{1}{7}$$

$$C = 2 (0)$$

37. (b) $x + y + 2z = a$

$$x + z = b$$

$$2x + y + 3z = C$$

$$R_2 \rightarrow R_1 \rightarrow R_2$$

$$R_3 \rightarrow 2R_1 \rightarrow R_3$$

$$\begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} a \\ a-b \\ 2a-c \end{bmatrix}$$

$$R_3 \rightarrow R_3 - R_2$$

$$\begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} a \\ a-b \\ a+b-c \end{bmatrix}$$

$$a+b-c = 0$$

$$a+b=c \quad (B)$$



38. (c) $f(x) = x^2 - bx + c$, b is an odd positive integer.

$f(x) = 0$ has prime number as roots and $b + c = 35$.

Let α, β be the roots of $x^2 - bx + c = 0$

Hence, $\alpha + \beta = b, \alpha \times \beta = c$

As $\alpha + \beta = b, b$ is an odd integer

Sum of two numbers is odd when one is even and other is odd number.

As we know α, β are prime numbers, so α must be 2 for given conditions.

$$f(2) = 4 - 2b + c = 0$$

$$2b - c = 4$$

$$b + c = 35$$

From Eqs. (i) and (ii)

$$b = 13, c = 22$$

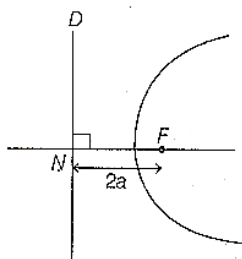
$$f(x) = x^2 - 13x + 22$$

$$f'(x) = 2x - 13 = 0 \Rightarrow x = \frac{13}{2}$$

There is only one critical point.

$$\text{It has } f_{\min} = f\left(\frac{13}{2}\right) = \left(\frac{13}{2}\right)^2 - 13 \times \left(\frac{13}{2}\right) + 22 = -\frac{81}{4}$$

39. (d) $F(-1, 1)$, directrix : $4x + 3y - 24 = 0$



$$FN = 2a$$

$$\left| \frac{4(-1) + 3(1) - 24}{\sqrt{4^2 + 3^2}} \right| = 2a$$

$$\left| \frac{-4 + 3 - 24}{5} \right| = 2a$$

$$\Rightarrow a = \frac{5}{2}$$

$$\text{Slope of directrix} = -\frac{4}{3}$$

$$\text{Slope of } FN = \frac{3}{4}$$

$$\text{Hence equation of } FN, y - 1 = \frac{3}{4}(x + 1)$$

Solving these two equations, we get N

$$x = 3, y = 4$$

$$\therefore N(3, 4)$$

Now, vertex is mid-point of F and N .

$$\therefore V(x, y) = \left(\frac{3 + (-1)}{2}, \frac{4 + 1}{2} \right) \Rightarrow V = \left(1, \frac{5}{2} \right)$$

40. $y = x^2 - x \sin x - \cos x = 0$

$$\frac{dy}{dx} = 2x - \sin x - x \cos x + \sin x = 0$$

$$2x - x \cos x = 0$$

$$x(2 - \cos x) = 0$$

$$\cos x = 2 \text{ (not possible)}$$

Number of solution are zero.

41. (a) Books on fairy tales can be arranged in $4!$ ways novels can be arranged in $5!$ ways.

Books on plays can be arranged in $3!$ ways.

Number of arrangement in which fairy tales books are arranged first, then novels are after that books on plays.

$$= 4! \times 5! \times 3! = 24 \times 120 \times 6 = 172$$

42. (c) $\sigma_x^2 = \frac{\sum d_i^2}{n}$

Here, the deviations are taken from the mean. Since A and B have both 100 consecutive integers, therefore both have same standard deviations and variance.

$$\therefore \frac{V_A}{V_B} = 1$$

(As $\sum d_i^2$ is same in both the case)

43. (c) Given, $|a + b| = \sqrt{29}$

$$\text{And } a \times (2\hat{i} + 3\hat{j} + 4\hat{k}) = (2\hat{i} + 3\hat{j} + 4\hat{k}) \times b$$

$$a \times (2\hat{i} + 3\hat{j} + 4\hat{k}) = -b \times (2\hat{i} + 3\hat{j} + 4\hat{k})$$

$$\Rightarrow (a + b) \times (2\hat{i} + 3\hat{j} + 4\hat{k}) = 0$$

$$\Rightarrow a + b = \lambda(2\hat{i} + 3\hat{j} + 4\hat{k}) \Rightarrow |a + b| = |\lambda| \sqrt{4 + 9 + 16}$$

$$\Rightarrow \sqrt{29} = |\lambda| \sqrt{29} \Rightarrow |\lambda| = 1 \Rightarrow \lambda = \pm 1$$

$$\therefore a + b = \pm(2\hat{i} + 3\hat{j} + 4\hat{k})$$

$$\text{Hence, } (a + b) \cdot (-7\hat{i} + 2\hat{j} + 3\hat{k})$$

$$\Rightarrow (2\hat{i} + 3\hat{j} + 4\hat{k}) \cdot (-7\hat{i} + 2\hat{j} + 3\hat{k}) \Rightarrow -14 + 6 + 12 = 4$$

44. (c) $x_1^2 + x_2^2 + x_3^2 + \dots + x_n^2 = 400$

$$x_1 + x_2 + x_3 + \dots + x_n = 80$$

AM of m^{th} power of terms $>$ m^{th} power of AM

$$= \frac{x_1^m + x_2^m + x_3^m + \dots + x_n^m}{n} > \left(\frac{x_1 + x_2 + x_3 + \dots + x_n}{n} \right)^m$$

$$m = 2$$

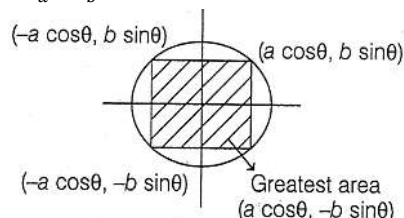
$$= \frac{x_1^2 + x_2^2 + x_3^2 + \dots + x_n^2}{n} > \left(\frac{x_1 + x_2 + x_3 + \dots + x_n}{n} \right)^2$$

$$= \frac{400}{n} > \frac{(80)^2}{n^2}$$

$$\Rightarrow n > \frac{6400}{400} \Rightarrow n > 16$$

So, 20 is answer.

45. (b) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$



$$\text{Area of rectangle} = l \times b$$

$$2a \cos \theta \times 2b \sin \theta$$

$$2ab \sin 2\theta$$

$$\text{For greatest } \sin 2\theta = 1 \Rightarrow (-1 \leq \sin \theta \leq 1) \Rightarrow \theta = \frac{\pi}{4}$$

$$\text{Area of greatest rectangle} = 2ab$$

46. The equation of tangent to $y^2 = 8ax$



$$y = mx + \frac{2a}{m} \quad (i)$$

$$\text{touches } x^2 + y^2 = 2a^2$$

$$\left(\frac{2a}{m}\right)^2 = 2a^2(1+m^2)$$

$$2 = m^0(m^2+1)$$

$$m^4 + m^2 - 2 = 0$$

$$(m^2+2)(m^2-1) = 0$$

$$m^2 - 1 = 0 \Rightarrow \pm 1$$

put in equation (1)

$$y = \pm (x+2a) \quad (B)$$

47. (a) $a_1, a_2, a_3, \dots, a_n$ are in AP (given)

$$a_1 = 0$$

$$\text{So, } a_2 = a_1 + d = d$$

$$a_3 = a_1 + 2d = 2d$$

$$a_{n-1} = (n-2)d \Rightarrow a_n = (n-1)d$$

$$\therefore \left(\frac{a_3}{a_2} + \frac{a_4}{a_3} + \dots + \frac{a_n}{a_{n-1}}\right) - a_2 \left(\frac{1}{a_2} + \frac{1}{a_3} + \dots + \frac{1}{a_{n-2}}\right)$$

$(n-2)$ terms $(n-3)$ terms

$$= \left[\left(\frac{2d}{d}\right) + \frac{3d}{2d} + \dots + \frac{(n-1)d}{(n-2)d}\right] - d_2 \left(\frac{1}{d} + \frac{1}{2d} + \dots + \frac{1}{(n-3)d}\right)$$

$$= \left[2 + \frac{3}{2} + \dots + \frac{(n-1)}{(n-2)}\right] - \left[1 + \frac{1}{2} + \dots + \frac{1}{n-3}\right]$$

$$= 1 + 1 + 1 \dots (n-2) \text{ times} + \left\{1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{(n-3)} + \frac{1}{(n-2)}\right\}$$

$$- \left\{1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{(n-3)}\right\}$$

$$= (n-2) + \frac{1}{(n-2)}$$

48. (a) $\cos 20^\circ + \cos 100^\circ + \cos 140^\circ$

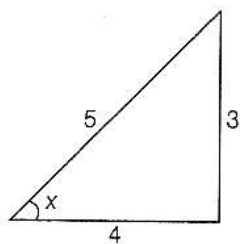
$$\Rightarrow 2\cos\left(\frac{100+20}{2}\right)\cos\left(\frac{100-20}{2}\right) + \cos 140^\circ$$

$$\Rightarrow 2\left(\frac{1}{2}\right)\cos 40^\circ + \cos(180^\circ - 40^\circ)$$

$$\Rightarrow \cos 40^\circ - \cos 40^\circ = 0$$

30. (d)

$$\tan x = \frac{-3}{4}, x \in IV \text{ (quad)}$$



$$\Rightarrow \sin x = -\frac{3}{5} \Rightarrow \cos x = \frac{4}{5}$$

$$\Rightarrow \sin 2x = 2\sin x \cos x = -2\left(\frac{3}{5}\right)\left(\frac{4}{5}\right) = \frac{-24}{25}$$

$$a + 2b = -6c \Rightarrow a + 2b + 6c = 0 \Rightarrow |a + 2b + 6c| = 0$$

49. (d) Lexicographic order is the order in which the words appear in the dictionary.

For the word before and after check starts from the left end.

Set is $\{a, b, c, d, e, f, g\}$

The word before bacdefg is agfedcb.

The word after bacdefg is bacdegf.

From the option, of all the given options, (a), (b), (c) are different from given order of word.

50. Foci of the ellipse are $(\pm\sqrt{16-b^2}, 0)$

Foci of hyperbola are $\left(\pm\sqrt{\left(\frac{12}{5}\right)^2 + \left(\frac{9}{5}\right)^2}, 0\right)$

If they coincide

$$16-b^2 = \frac{144+81}{25}$$

$$b^2 = 7 \quad (b)$$

REASONING SOLUTION

51. (c) Total number of fruits in the basket

$$= 19 \times 1 - 4 \times 2$$

$$= 19 - 8 = 11$$

52. (d) Total number of two's = total number of orange = 6

Total number of four = 4

$\therefore$ Total number of oranges in the basket

$$= 6 - 4 = 2$$

Solutions (Q. Nos. 53- 56)

From the given information the order will be as

8	K
7	N
6	J
5	Q
4	L
3	O
2	M
1	P

53. (a) After interchanging the places, O lives between P and M.

54. (b) Only two people live between M and Q.

55. (c) K lives on floor number 8.

56. (d) Only O and M are adjacent to each other.



Solutions (Q. Nos. 57- 59)

Forwards - A, C, E and J
Point Guard - B, G and H
Defenders - D, F and I

57. (a) The largest possible team shall be of 6.
F, J, A, D, G and B/E.

58. (b) H cannot be included in 6 member team as if H is included then A and D will not be in the team.

59. (c) G will be a member of 6 member team.

60. (d) Through option we can check the possibilities. With option (d), we have

B - B - B - B

S - S - S

So, each boy has 3 sisters and 3 brothers and each girl has 2 sisters and 4 brothers.

61. (c)



As, good = 4

Picture = 7

So, see = 8

62. (c) As, $E \rightarrow 05 \rightarrow 0 + 5 = 5$
 $X \rightarrow 24 \rightarrow 2 + 4 = 6$
 $A \rightarrow 01 \rightarrow 0 + 1 = 1$

M $\rightarrow 13 \rightarrow 1 + 3 = 4$

I $\rightarrow 09 \rightarrow 0 + 9 = 9$

N $\rightarrow 14 \rightarrow 1 + 4 = 5$

A $\rightarrow 01 \rightarrow 0 + 1 = 1$

T $\rightarrow 20 \rightarrow 2 + 0 = 2$

I $\rightarrow 09 \rightarrow 0 + 9 = 9$

O $\rightarrow 15 \rightarrow 1 + 5 = 6$

N $\rightarrow 14 \rightarrow 1 + 4 = 5$

G $\rightarrow 07 \rightarrow 0 + 7 = 7$

Similarly,

O $\rightarrow 15 \rightarrow 1 + 5 = 6$

V $\rightarrow 22 \rightarrow 2 + 2 = 4$

E $\rightarrow 05 \rightarrow 0 + 5 = 5$

R $\rightarrow 18 \rightarrow 1 + 8 = 9$

N $\rightarrow 14 \rightarrow 1 + 4 = 5$

M $\rightarrow 13 \rightarrow 1 + 3 = 4$

E $\rightarrow 05 \rightarrow 0 + 5 = 5$

N $\rightarrow 14 \rightarrow 1 + 4 = 5$

T $\rightarrow 20 \rightarrow 2 + 0 = 2$

63. (d) None of the given statements match the facts.

64. (d) There are 19 numbers which has 6 as a digit when unit digit 6 is replaced by 9, then sum of numbers will be increase by $= 3 \times 10 = 30$

From 60 to 69 there are 10 numbers which have ten's digit 6.

Here, the increase will be $30 \times 10 = 300$

$\therefore$ Total increase $= 30 + 300 = 330$

65. (a) Let the number of children be x.

$$\therefore x \times \frac{x}{5} = 405$$

$$\Rightarrow x^2 = 5 \times 405 = 2025$$

$$\Rightarrow x = \sqrt{2025} = 45$$

66. (c) Number of terms in

$$\text{1st series} \quad 817 = 17 + (m - 1)4$$

$$\Rightarrow m = 201$$

$$\text{2nd series,} \quad 851 = 16 + (n - 1)5$$

$$\Rightarrow n = 168$$

$$\text{Now, } 17 + (m - 1)4 = 16 + (n - 1)5$$

$$\Rightarrow 4m + 2 = 5n$$

$$\text{For, } m = 2, \quad n = 2$$

$$m = 7, \quad n = 6$$

$$m = 12, \quad n = 10$$

$$m = 17, \quad n = 14$$

It also form a series.

$\therefore$ Number of common terms.

$$197 = 2 + (P - 1) 5$$

$$P = 40$$

67. (b) (a) ONGEAR - ORANGE

(b) NOONI - ONION

(c) ALPEP - APPLE

(d) AUVAG - GUAVA

Onion is the only thing that is taken out of the soil.

68. (c) At C, number of passenger = 55

$$\therefore \text{Total number of passenger at B} = \frac{5}{4} \times (55 - 3) \\ = \frac{5}{4} \times 52 = 65$$

$\therefore$ Total number of passengers at

$$A = \frac{6}{5} \times (65 - 10)$$

$$= \frac{6}{5} \times 55 = 66$$

69. (a) If the answer of question is 'No', then none belongs No-zone. This implies both are Yes. Which is contradiction.

If Kumar is 'Yes' the answer to the questions is 'Yes' that implies that Kevin is from No.

70. (a) Raman's DOB = 5th March, 1970

$\therefore$ Lakshman's DOB = 8th Feb 1970

Republic Day, i.e. 26 Jan = Monday

$\therefore$ 8th Feb = Sunday.

GOVERNMENT-7645954552



71. (c) If P is true, then Q is true.
If Q is true, then R is true.
So, if P is true, then Q and R are both true.
and if Q and R are true, then S is false.

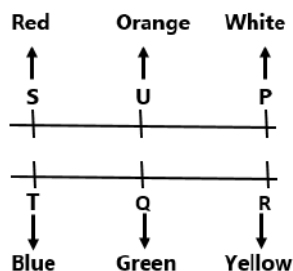
72. (b)
-
- So, C is son of A and E.

73. (b) Total hours from 5 am to 10 pm of 3rd day = 65h
The clock loses 16 min in 24 h.
i.e. 23 h 44 min of the clock = 24h of the correct clock
So, 65 h of clock = $\frac{24 \times 15 \times 65}{356}$ of the correct clock
= 65.73h
= 65h 44 min
So, correct time = 10:45pm (approx)

74. (b) 4 days after Monday is Friday
So, today is Tuesday.

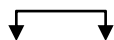
Solutions. (Q. Nos. 75-77)

From the given information the arrangement will be as follows.



Order of height $T > S/Q > P > R > U$

75. (d) 2nd highest cannot be determined.
76. (b) 2nd shortest house is R.
77. (b) The tallest house is T which is Blue.



78. (a)
NECESSARY
There is only one such pair NS.

79. (a) The series is as follows

4

7

$$7 + 4 = 11$$

$$11 + 7 = 18$$

$$11 + 18 = 29$$

$$18 + 29 = 47$$

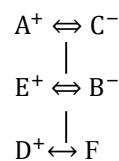
$$29 + 47 = 76$$

$$47 + 76 = 123$$

$$76 + 123 = 199$$

∴ Missing term = 76

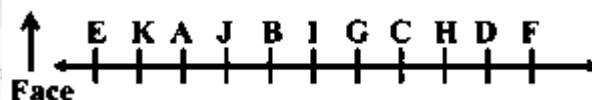
Solutions (Q. Nos. 80-82) From the given information,



According to weight $A > C > E > B > D, F$

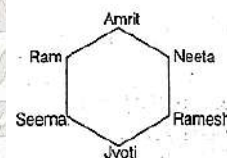
80. (d) BE is a couple
81. (a) C will be at 2nd place.
82. (a) C is grandmother of D.

Solutions (Q. Nos. 83-85) From the given information the arrangement will be as



83. (d) I is sitting in the middle of the row.
84. (c) D and K will be at end points.
85. (b) CHDF are sitting to the right of G.

Sol. (Q.Nos. 86-89) From the given information the arrangement is as follows.



86. (a) After interchanging their position Jyoti will be opposite to Neeta.
87. (d) Ram is sitting left to Amrit.
88. (a) Neeta is sitting between Amrit and Ramesh.
89. (c) Amrit is sitting opposite to Jyoti.
90. (d) We can check it through option.

ENGLISH

91.	b	92.	d	93.	a	94.	a	95.	d
96.	b	97.	c	98.	b	99.	a	100.	c
101.	d	102.	b	103.	c	104.	a	105.	b
106.	b	107.	d	108.	a	109.	b	110.	a & c

COMPUTER

111. (a)

42797 KB will take 85512 sectors (42797×1024 bytes/512 bytes)
Since, there are 64 sectors per surface $85512/64 = 1337.406$
sectors are required, so we take 1338 sectors. These sectors are distributed among 16 surfaces, so $1338/16 = 83.58$ cylinders will be required. So the final answer will be $84 + 1200 = 1284$.
One more fact to be noted is that the file occupies 83.58 cylinders, but the 0.58 cannot be accommodated in the first one (the files storage starts from $< 1200, 9, 40 >$)
Hence, the file will be extended to 198 (85594) more bytes of cylinder 1284.



112. (b)

K-map for $F(P, Q, R, S) = \sum(0, 2, 5, 7, 8, 10, 13, 15)$ where 2, 7, 8, 13 are do not care terms

	RS	RS	RS	RS
PQ	1	0	1	3
PQ	4	5	7	6
PQ	12	13	15	14
PQ	X	8	9	11

There are 2 quads.

Quad 1 ($m_0 + m_2 + m_8 + m_{10}$) = $\bar{Q} \bar{S}$

Quad 2 ($m_5 + m_7 + m_{13} + m_{15}$) = $Q S$

Hence, minimal SOP form = $QS + \bar{Q} \bar{S}$

113. (c)

The given Venn diagram represent XNOR expression.

Hence, $a \text{ XNOR } b = ab + a'b'$

114. (b)

The range of values for n-bit signed magnitude representation is $-(2^{n-1} - 1)$ to $(2^{n-1} - 1)$.

115. (a)

We know that,

$$\text{Average latency} = \frac{\text{Max Latency}}{2}$$

$$\text{Where, Max latency} = \frac{1}{\text{rev/time}} = \frac{1}{\frac{6000}{60} \text{ S}}$$

$$= \frac{60}{6000} \text{ S} = \frac{1}{100} \text{ S}$$

$$\text{So the average latency} = \frac{10^{-2}}{2}$$

$$= 0.5 \times 10^{-2}$$

$$= 5 \times 10^{-3} \text{ s}$$

116. (d)

8 bit representation of 100 $\rightarrow 01100100$

Complement $\rightarrow 10011011$

Add 1 $\rightarrow 00000001$

-1 $\rightarrow 10011100$

Hence 2's complement representation is 10011100.

117. (*)

None option is correct.

118. (b)

119. (c)

Given, $(43)_8 = (y3)_8$

By converting to decimal system

$$4x + 3x^0 = y \cdot 8^1 + 3 \cdot 8^0$$

$$4x + 3 = 8y + 3$$

$$4x = 8y$$

$$x = 2y$$

x is the base of number system $x > 4$.

Y is number in the number system with base 8, so $y < 8$.

x	y
6	3
8	4
10	5
12	6
14	7

Number of possible solution = 5

120. (d)

$$(1101)_2 - (1010)_2$$

1's complement of 1010 is 0101.

Minued -1101

1's complement of subtrahend -0101

Carry over 10010

1

0011

The required difference is 0011.



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